
CF05: COMPLETE FORMALISM — INTEGRABILITY CLOSURE (NO HIDDEN CONSERVED QUANTITIES)

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Abstract

Metriplectic (GENERIC) dynamics couples a reversible Poisson flow to an irreversible metric flow. The VDM axiom-core discipline requires that (i) the Poisson operator satisfies Jacobi, (ii) the degeneracy constraints enforce energy conservation and non-negative entropy production, and (iii) no *unlisted* conserved quantities exist that would silently constrain dynamics or obstruct relaxation. This CF formalizes an *integrability-closure* protocol: for a *specified* finite-dimensional truncation or discretization (fixed L, M, E, S), we distinguish structural invariants forced by **VDM-E-140–VDM-E-143** from accidental invariants, and we define a falsifiable gate suite that searches for additional independent first integrals across declared function classes. Analytic components include Darboux theory (polynomial/rational first integrals), Prolle–Singer-type searches (elementary/Liouvillian first integrals), and Kovalevskaya–Painlevé indicators (movable singularities). A numerical backstop performs regression-based invariant discovery on trajectories and then validates any candidates by long-horizon drift tests. The companion CFN notebook provides minimal Poisson/Jacobi and Hamiltonian-invariance sanity checks and serves as a scaffold for automation of the full closure certificate.

Keywords integrability · first integrals · Darboux theory · Prolle–Singer · Painlevé analysis · metriplectic · GENERIC · Poisson bracket · Void Dynamics Model

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1 Objective and scope

This CF defines a *closure discipline* for conserved quantities in VDM metriplectic (GENERIC) models. The motivating problem is simple: if a model carries *additional* independent first integrals beyond the invariants explicitly intended by the axioms and registries, then long-time behavior can be silently constrained. In practice this can: (i) obstruct relaxation even when the entropy law is satisfied, (ii) create spurious “superselection” leaves, and (iii) invalidate downstream empirical tests by changing the effective degrees of freedom.

Canon alignment. This document is written to the VDM canon conventions: axioms and gates [Lietz, 2025a], equation registry anchors [Lietz, 2025b], validation metrics [Lietz, 2025c], symbols [Lietz, 2025d], and provenance conventions [Lietz, 2025e].

Scope classification. This CF is **axiom-core** with respect to the metriplectic split (A4) and entropy law (A5), because “hidden invariants” are a direct threat to falsifiability. It is also **derived-limit** in its integrability tooling: Darboux/Prelle–Singer/Painlevé analyses are imported as *methods* for discovering or ruling out first integrals within declared function classes, not as foundational physics.

What this CF does and does not claim.

- **Does:** (i) formalize the distinction between structural invariants required by GENERIC and accidental invariants; (ii) define an implementable, falsifiable gate suite that searches for accidental invariants; (iii) provide a machine-readable “closure certificate” schema for publishing null results.
- **Does not:** prove global non-existence of extra invariants for *all* possible (L, M, E, S) choices. Any “no-hidden-integrals” statement must be attached to a *specific* model and a *declared* search class (e.g., polynomials up to degree d , Darbouxian functions with bounded degree, etc.).

2 Claim inventory and decisive falsifiers

2.1 Primary claims

Claim: C1: Structural invariants in GENERIC *Decisive metric: identity residuals for $\dot{E} = 0$, $\dot{S} \geq 0$, and degeneracy $L\nabla S = 0$, $M\nabla E = 0$.*

For a state x , energy $E(x)$, entropy $S(x)$, and operators (L, M) satisfying **VDM-E-140–VDM-E-143**, the model has:

$$\frac{dE}{dt} = 0, \tag{1}$$

$$\frac{dS}{dt} = \nabla S^\top M \nabla S \geq 0, \tag{2}$$

and the degeneracy constraints **VDM-E-142** enforce that S is a Casimir of the Poisson bracket and E is a Casimir of the metric bracket.

Claim: C2: Hidden invariants are model-specific *Decisive metric: closure search returns either a new independent integral or an explicit null result over a declared class.*

Beyond the structural invariants in C1, any additional first integrals or Casimirs arise from extra symmetries, constraints, or accidental algebraic structure. Therefore, “no hidden invariants” must be established by *search + verification* for the particular (L, M, E, S) used in a runner.

Claim: C3: Closure certificate is publishable *Decisive metric: certificate contains bounded search parameters + verification residuals + provenance.*

A published closure claim must include: (i) the exact model definition (hash/commit), (ii) which function classes were searched and to what bounds, (iii) a multi-seed verification procedure, and (iv) quantitative residuals. This CF defines such a certificate (§8).

2.2 Decisive gates

The following gates are designed to be binary and automatable. They intentionally separate *structure* (Poisson/GENERIC correctness) from *closure* (no extra invariants found).

Gate: G0: Poisson/Jacobi validity *Threshold: $e_{\text{Jacobi}} \leq 10 \varepsilon_{\text{mach}}$ on a declared basis.*

Compute the Jacobi residual e_{Jacobi} as defined in **VDM-E-141** (basis-restricted) and KPI `kpi-poisson-jacobi-resid`. If Jacobi fails, integrability statements are not meaningful because “Casimir” is ill-defined.

Gate: G1: Degeneracy correctness *Threshold: $\|L\nabla S\|_{\infty} \leq 10 \varepsilon_{\text{mach}}$ and $\|M\nabla E\|_{\infty} \leq 10 \varepsilon_{\text{mach}}$.*

Enforce **VDM-E-142** (KPI `kpi-degeneracy-resid`). Failure indicates a contradiction to the axioms (A4/A5).

Gate: G2: Entropy production nonnegativity *Threshold: per-step $\Delta S \geq -10 \varepsilon_{\text{mach}}$ and cumulative $\Delta S \geq -10 \varepsilon_{\text{mach}}$.*

Enforce **VDM-E-143** (KPI `kpi-entropy-prod-nonneg`).

Gate: G3: Hidden-invariant discovery gate *Threshold: No additional independent invariant passes verification over the declared search class.*

Run the analytic + numeric discovery searches in §5 and §6. If a candidate invariant $I(x)$ is discovered and passes G4 (drift), then closure fails. If no candidate passes, closure is certified *for that search class*.

Gate: G4: Candidate invariant drift *Threshold: relative drift $\leq 10^{-8}$ over T and stable across seeds.*

For any candidate I , define

$$\text{drift}(I) := \max_{t \in [0, T]} \frac{|I(x(t)) - I(x(0))|}{\max(1, |I(x(0))|)}. \quad (3)$$

A candidate passes only if $\text{drift}(I) \leq 10^{-8}$ across a declared set of seeds and step-size refinements.

2.3 Companion notebook outputs

The companion CFN notebook `Derivation/Notebooks/03_Closure/CF5_Integrability_Closure.ipynb` provides minimal sanity checks for G0 and for Hamiltonian invariance under a known symplectic rotation. It is intentionally *not* the full closure harness. This CF specifies the additional outputs required to complete the closure automation:

- a Darboux search report (degree cutoffs, candidates, cofactors, independence tests),
- a Prelle–Singer-style integrating factor search log (ansatz family and bounds),
- a Painlevé/Kovalevskaya resonance table for declared reductions,
- a trajectory-based invariant regression report and drift plots.

3 Foundations: first integrals and metriplectic structure

3.1 First integrals and functional independence

Definition 1 (First integral). Let $\dot{x} = F(x)$ be an autonomous ODE on a domain $\mathcal{X} \subset \mathbb{R}^n$. A differentiable scalar function $I : \mathcal{X} \rightarrow \mathbb{R}$ is a *first integral* if

$$\frac{d}{dt} I(x(t)) = \nabla I(x)^\top F(x) = 0 \quad \text{along all solutions.} \quad (4)$$

Two integrals I_1, I_2 are *functionally independent* on a set if their gradients are generically linearly independent. In Hamiltonian systems, complete integrability is often defined via *Liouville integrability* (n degrees of freedom requires n integrals in involution) [Arnold, 1989]. In dissipative metriplectic models, the relevant closure question is weaker: *do any unintended invariants exist that constrain the full flow or the Poisson leaves?*

3.2 GENERIC / metriplectic evolution

VDM uses the GENERIC (metriplectic) split [Morrison, 1986, Grmela and Öttinger, 1997, Öttinger and Grmela, 1997, Öttinger, 2005]:

$$\dot{x} = L(x) \nabla E(x) + M(x) \nabla S(x), \quad (5)$$

with antisymmetric Poisson operator $L^\top = -L$ and symmetric positive semidefinite metric operator $M^\top = M \succeq 0$ (**VDM-E-140**). The Poisson bracket of functionals is $\{F, G\}_J = \nabla F^\top L \nabla G$ (**VDM-E-141**).

The degeneracy identities (**VDM-E-142**) are

$$L \nabla S = 0, \quad M \nabla E = 0, \quad (6)$$

and imply:

$$\frac{dE}{dt} = \nabla E^\top L \nabla E + \nabla E^\top M \nabla S = 0, \quad (7)$$

$$\frac{dS}{dt} = \nabla S^\top L \nabla E + \nabla S^\top M \nabla S = \nabla S^\top M \nabla S \geq 0. \quad (8)$$

Equation (8) is the entropy production law **VDM-E-143**.

Remark 1 (Structural vs accidental invariants). Equations (7)–(8) are *structural*. They hold for any (L, M, E, S) satisfying (5)–(6). Any additional first integrals are *accidental* in the sense that they require extra symmetry, constraints, or special algebraic form.

4 What counts as a “hidden invariant”?

The closure problem is not limited to conserved scalars of the full flow. In metriplectic models there are three common “places” where unintended invariants can hide:

1. **Full-flow first integrals:** I such that $\nabla I \cdot (L \nabla E + M \nabla S) = 0$ identically. These constrain the long-time attractor.
2. **Poisson Casimirs:** C such that $L \nabla C = 0$. Even if not conserved under the full flow, Casimirs define invariant leaves for the reversible limb and strongly shape dynamics.
3. **Metric null invariants:** J such that $M \nabla J = 0$. These are invariants of the irreversible limb and can prevent entropy maximization.

The GENERIC axioms guarantee E is in the metric nullspace and S is in the Poisson nullspace. Closure is the assertion that there are *no additional independent quantities of these types* beyond those explicitly listed.

5 Analytic search components

Analytic methods are used here in a pragmatic way: *if* a hidden integral exists in a simple function class, these tools can often find it. If they find nothing, the result is a *bounded null result* that becomes publishable once paired with numerical verification.

5.1 Darboux theory for polynomial/rational first integrals

Assume a polynomial vector field $\dot{x} = F(x)$ with components $F_i \in \mathbb{R}[x_1, \dots, x_n]$.

Definition 2 (Darboux polynomial). A non-constant polynomial $f \in \mathbb{R}[x]$ is a *Darboux polynomial* if there exists a polynomial $K \in \mathbb{R}[x]$ (the *cofactor*) such that

$$\nabla f(x)^\top F(x) = K(x) f(x). \quad (9)$$

If a set $\{f_i\}$ of Darboux polynomials share compatible cofactors, one can form first integrals of Darbouxian type

$$I(x) = \prod_i f_i(x)^{\lambda_i} \quad (10)$$

with exponents λ_i chosen so that the cofactors cancel in the logarithmic derivative. This is classical Darboux theory [Christopher et al., 2002].

Gate implication. A practical closure search enumerates Darboux polynomials up to degree d and tests whether any nontrivial (10) is functionally independent of E and the listed Casimirs. Discovery of such an I triggers **G3 FAIL**.

5.2 Prelle–Singer searches for elementary/Liouvillian first integrals

For rational vector fields, the Prelle–Singer theory establishes that the existence of an *elementary* first integral implies the existence of an integrating factor of restricted form [Prelle and Singer, 1983]. In its original setting (first-order ODEs), the method searches for an integrating factor μ such that

$$\mu(x) (F(x) \cdot dx) \quad (11)$$

is an exact differential. Many extensions exist (Liouvillian/generalized integrating factors), but the closure use is the same: search within a declared ansatz family (bounded degrees/exponents) and report either a found integral or a null result.

Gate implication. If an elementary/Liouvillian first integral is found within the declared ansatz bounds and passes drift verification (G4), closure fails.

5.3 Kovalevskaya–Painlevé indicators

Painlevé analysis studies movable singularities of solutions. A classical heuristic is: if general solutions exhibit movable branch points or essential singularities, the system is unlikely to be integrable in terms of meromorphic functions. The Ablowitz–Ramani–Segur program relates nonlinear evolution equations to Painlevé-type ODE reductions [Ablowitz et al., 1980a,b].

Use as closure evidence (not a proof). In this CF, Painlevé/Kovalevskaya computations are treated as an *indicator* that can quickly falsify naive “integrable” expectations. A failure does not mathematically prove non-integrability in all senses, but it is strong evidence against hidden analytic first integrals of simple type.

6 Numerical discovery and verification

When symbolic methods are inconclusive or too expensive, the closure harness uses trajectory data to propose invariants and then tests them across seeds and step refinements.

6.1 Data-driven invariant regression

Let $x(t_k)$ be sampled from a numerical integrator for $\dot{x} = F(x)$. Choose a candidate function class $\mathcal{H}$ spanned by basis functions $\phi_1, \dots, \phi_m$ (e.g., monomials up to degree d). Seek $I(x) = \sum_j c_j \phi_j(x)$ such that

$$\nabla I(x(t_k))^\top F(x(t_k)) \approx 0 \quad (12)$$

in least squares. Equivalently, estimate $\dot{I}(t_k) \approx (I(t_{k+1}) - I(t_k))/\Delta t$ and minimize $\|\dot{I}\|$.

Independence filter. A candidate is rejected if it is functionally dependent on known invariants. Practically, one checks rank of the gradient matrix $[\nabla E, \nabla C_1, \dots, \nabla I]$ on sampled points or uses regression/correlation diagnostics.

6.2 Drift verification (G4)

Candidate invariants are only admitted if they survive long-horizon integration:

- multiple seeds and multiple step sizes,
- both J-only, M-only, and J $\oplus$ M compositions when relevant,
- drift thresholds as specified in G4.

This step is essential: low $\dot{I}$ on a short trajectory can be an artifact of basis choice, stiffness, or numerical error.

7 Worked example: minimal Poisson/Jacobi sanity checks

The companion CFN notebook implements a minimal 2D canonical bracket check using test polynomials and finite-difference approximations of nested brackets. For a single random point (q, p) , it reports residuals (machine-specific):

Check	Residual
Poisson antisymmetry residual	0
Poisson bilinearity residual	5.55e−17
Jacobi residual (finite-difference nested brackets)	2.32e−10
Energy drift under exact symplectic rotation (N steps)	5.94e−15

Table 1: Notebook sanity-check residuals for the canonical 2D Poisson bracket and a harmonic oscillator rotation map. The Jacobi residual is dominated by finite-difference error; analytic evaluation yields exact satisfaction for constant L .

Interpretation. The canonical constant Poisson operator satisfies Jacobi exactly. The point of this worked example is not to certify production thresholds (which require analytic or symbolically exact bracket evaluation), but to provide a simple, falsifiable scaffold demonstrating how a Jacobi residual meter is computed and reported.

8 Machine-readable closure certificate

A closure claim should be publishable as a small JSON sidecar alongside the PDF. This CF recommends the following minimal schema:

```
{ "model_id": "<name>", "commit": "<git sha>", "operators": { "L": "<hash>", "M":
"<hash>" }, "functionals": { "E": "<hash>", "S": "<hash>" }, "search_classes": [
{ "method": "darboux", "max_degree": d }, { "method": "prelle_singer", "ansatz":
"<desc>", "bounds": "<desc>" }, { "method": "poly_regression", "max_degree": d }
], "seeds": [ ... ], "gates": { "G0": <value>, "G1": <value>, "G2": <value>,
"G3": "PASS", "G4": <value> }, "discovered_invariants": [ ... ], "notes": "<any
caveats>" }
```

9 Gate suite summary

For publication friendliness, we restate the closure gates in one place:

- **G0 (Poisson/Jacobi):** Jacobi residual e_{Jacobi} within machine-epsilon scaling.
- **G1 (Degeneracy):** $\|L\nabla S\|_\infty$ and $\|M\nabla E\|_\infty$ within machine-epsilon scaling.
- **G2 (Entropy production):** per-step and cumulative $\Delta S \geq 0$ to tolerance.
- **G3 (No hidden invariants):** no additional independent invariant survives discovery + verification over declared classes.
- **G4 (Drift):** any candidate invariant must exhibit $\leq 10^{-8}$ relative drift over long horizons and be stable across seeds/step refinements.

10 Integration map

Downstream use in VDM. This CF is intended as a *pre-flight check* for any new metriplectic variable set or truncation. Before running expensive experiments, one should: (i) certify G0–G2 (structure), then (ii) generate a closure certificate (G3–G4) for the declared search classes.

Relationship to other CF modules.

- CF01–CF02 provide geometric/construction origins for (L, M) and the degeneracy structure.

- CF03–CF04 supply additional axiom-core constraints (hierarchy and causality) that can be violated by accidental invariants.
- CF05 provides the “no unintended constraints” check that stabilizes interpretation of downstream dynamical tests.

11 Provenance and reproducibility

Commit: 91332f4cf89a52d239eaab6659b8df8d8b9f3202 **Seed(s):** N/A **Artifacts:**
Source: Derivation/Complete-Formalisms/CF05_Integrability_Closure.md (SHA256:
 8890add6beb6271c5a7ad15b7fb3ca0d9441ee53a3cc9f7adb229df27f737ac3)

12 Limitations and future work

- **Bounded classes only:** closure is only certified over the declared search families (degree cutoffs, ansatz bounds). Stronger claims require expanding those bounds.
- **High-dimensional and field settings:** infinite-dimensional PDEs can carry families of Casimirs; closure should be performed on each discretization/truncation actually used.
- **Painlevé indicators are not proofs:** they are used as fast falsifiers, not as rigorous non-integrability theorems.
- **Automation:** the current CFN notebook is a scaffold; a production closure meter should use analytic nested-bracket evaluation to meet the Jacobi thresholds in [Lietz, 2025c].

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